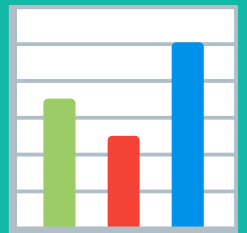


Customer Churn & Retention Analysis

A Data-Driven Approach Using Python, SQL, and Power BI

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Data Preparation & SQL Analytics

Transforming raw customer data into analysis-ready datasets



Data Cleaning Steps

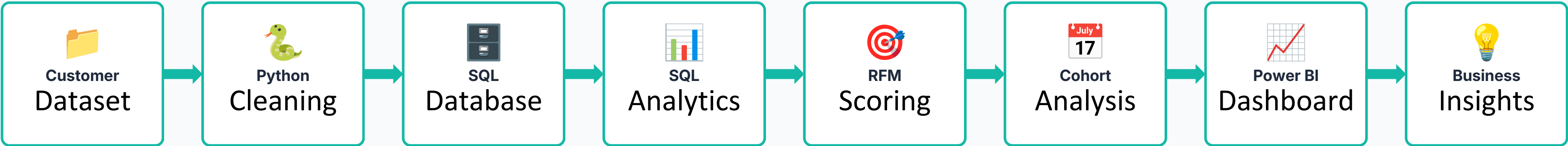
- Median imputation for missing values
- Standardize categorical values
- Check invalid values and outliers
- Validate data consistency



SQL Analytics Approach

- Store cleaned data in relational database
- CTEs for structured analysis
- Aggregations for customer-level metrics
- NTILE for RFM scoring
- DENSE_RANK for ranking

PROJECT WORKFLOW



RFM Customer Segmentation

Quintile-based scoring to classify customers into actionable segments



Recency
How recently did the customer purchase?



Frequency
How often does the customer purchase?



Monetary
How much value/spending is associated?

CUSTOMER SEGMENTS



Champions
High recency + frequency + value



Loyal Customers
Consistent customers



Potential Loyalists
Recent customers with potential

KEY SEGMENT



At Risk
Previously valuable, showing inactivity



Lost / Hibernating
Inactive with low engagement

Cohort & Retention Analysis

Understanding customer lifecycle and retention patterns

What We Study

- July 17

Tenure Periods

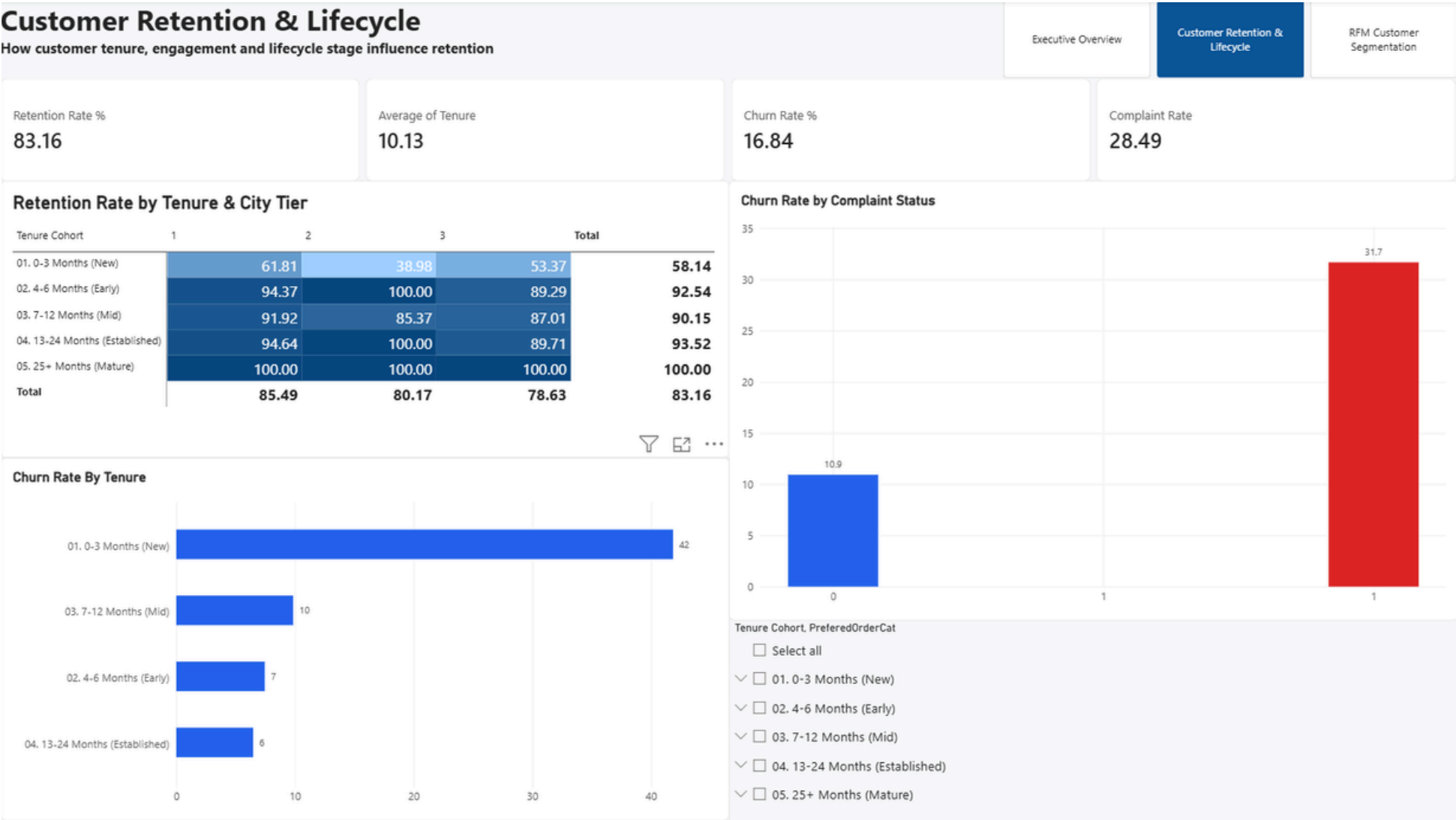
How long has the customer been with us?
- Customer Groups

Comparing behavior across segments
- Geographic Tiers

Regional differences in retention

Key Question

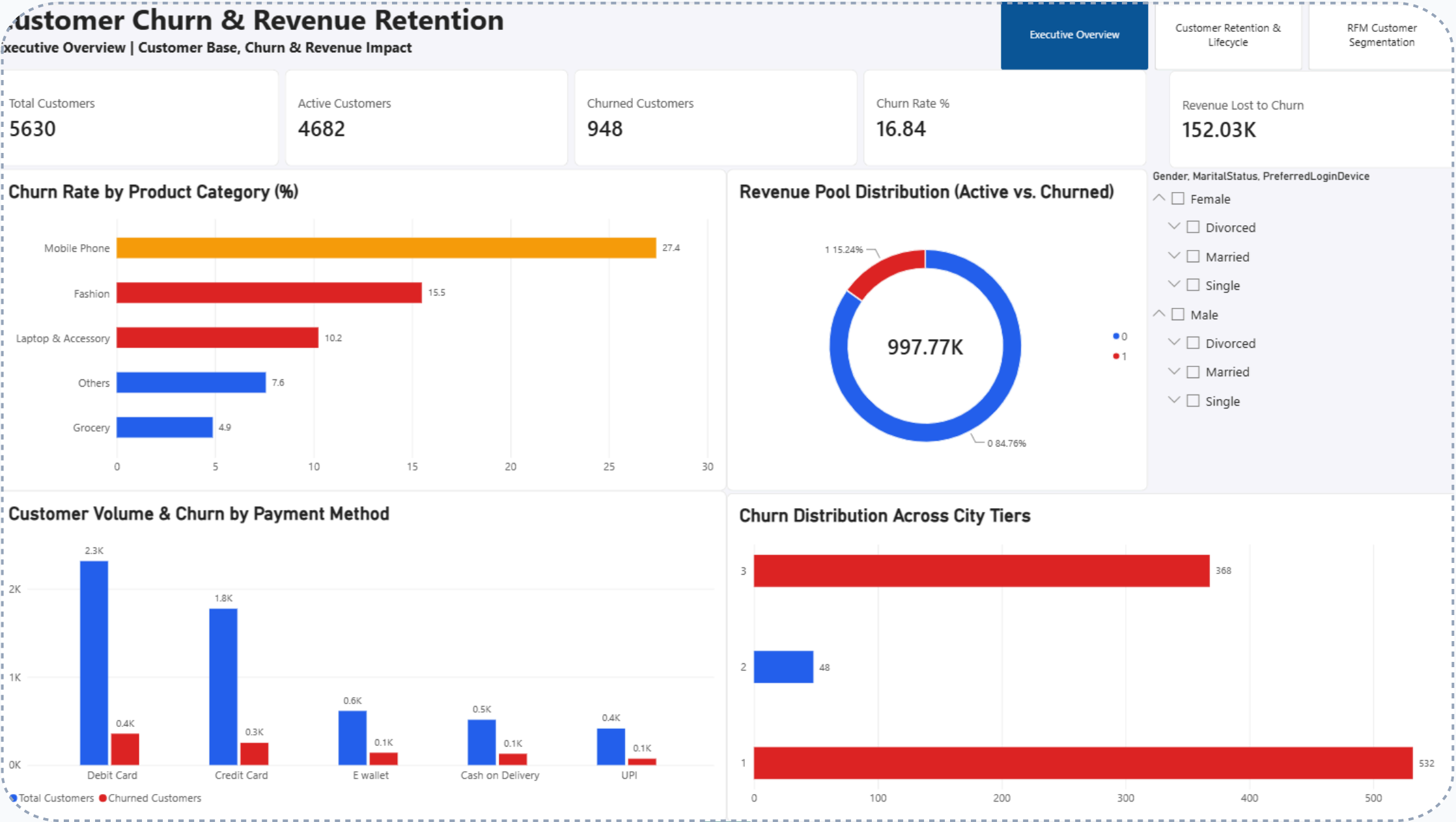
At which lifecycle stage are customers most likely to become inactive?



Power BI Dashboard

Business users can explore customer behavior without writing code

-  Interactive KPIs
-  Churn Analysis
-  Customer Segmentation
-  Retention & Cohort Views
-  Customer-Level Details
-  Dynamic Slicers & Filters
-  DAX-based Calculations
-  Drillthrough



Key Results & Business Insights

What the analysis tells us about customer behavior



AT-RISK SPENDING EXPOSURE

\$400,000+

in spending from customers at risk of churning



High-Value at Risk

High-value customers can exist
within the At-Risk segment



Recency & Tenure

Customer inactivity examined
through recency and tenure



RFM Prioritization

RFM segmentation helps
prioritize retention efforts



Lifecycle Patterns

Cohort analysis shows how retention
changes over the lifecycle



Interactive Exploration

Power BI makes results
easy to explore interactively

Feasibility, Limitations & Future Scope

Feasibility

Technical

Uses standard Python, SQL,
and Power BI tools

Economic

Mostly open-source tools,
standard computer hardware

Operational

Dashboard usable by
non-technical business users

Current Limitations

- Current segmentation is rule-based, not ML-based
- Analysis depends on the available dataset
- Historical data may not capture every reason for customer churn

Future Extensions

- Logistic Regression
- Random Forest
- XGBoost
- Individual churn probability
- SHAP explainability
- Automated retention recommendations
- Automated win-back campaigns

** These are planned extensions, not yet implemented*

Conclusion



This project uses customer data to identify churn patterns and find which customer groups need attention the most.

Thank You